Data Dictionary

In the SAP API Hub, in the **Consumer API to Request for Planning in External System**, you can view the field details in the **Model View**. In the **API Reference** section, you can view the example values.

In the downloaded YAML file you can view the following details of the fields in the API interface:

- Field name
- Type
- Description
- Max length
- Required or not
- Example

Relevant Codes

SAP Business Network for Logistics has defined its own code values for the various attributes used within business network. These codes are used during B2B communication so using APIs in business network. For more details on the list of codes supported in SAP Business Network for Logistics, see Code Lists.

For planning in external system, the following code lists are relevant:

- **Document Reference Type Codes**

  As of now, only code 1122 - Transportation Order is supported.

- **Contract Condition Codes**
- **Item Category Code**
- **Movement Types**
- **Shipping Type Descriptions**

  As of now, the following shipping type codes are supported:

Code	Description
17	LTL (Less Than Truck Load)
18	FTL (Full Truck Load)
4	Bulk

- **Transportation Document Category Code**

  As of now, the following document category codes are supported:

Code	Description
BO	Booking
SO	Service Order

Code	Description
TO	Transportation Order (Freight Order)
TU	Transportation Unit (Different from Freight Unit)
SA	Shipment Advice
FU	Freight Unit
SH	Shipment

- **Transportation Document Type Code**

  As of now, the following document type codes are supported:

Code	Description
1	`TRANSPORTATION_REQUEST`
2	`TRANSPORTATION_ORDER`

- **Transport Means Standard Code Descriptions**

  As of now, the following transport means standard codes are supported:

Code	Description
8	Container ship
11	Ship
13	Ocean vessel
16	Barge
28	Refrigerated truck and trailer
29	Freezer truck and trailer
31	Truck
41	Freezer truck
42	Isothermic truck
43	Refrigerated truck
46	Refrigerated van
65	Container only vessel

- **Unit of Measure Temperature**
- **Unit of Measure Volume**
- **Unit of Measure Weight**

Additional Information

- **FUR Item**

  You can skip the items with the category code as FUR.

- **Business-to-Business Routing**

  This consumer API supports business-to-business routing. The system uses the forwarding agent or carrier information that the shipper has specified in the sales order in their transportation system to route the information to the forwarding agent or the carrier. If this information is unavailable, then the system uses the network partner information that the shipper has specified in the **Configure Partner Connections** app.

- **Cancel Scenario**

  When you cancel a freight unit that already has a planned freight document in your external transportation system, the system cancels the freight unit and the link to the freight document is removed in your external transportation system. Moreover, this freight document will no longer have the items linked to that freight unit and the status becomes In Process.

Provider API to Create Planned Freight Document from External System

As a carrier, you can use this provider API to create a planned freight document in the SAP Business Network for Logistics system based on the planning activities that you performed in your external system.

You can also use this API to receive a request for cancelation of a freight document in your SAP Business Network for Logistics system from an external system.

Description

SAP Business Network for Logistics system.

Connection Information

Technical Name	Field
Company Type	Carrier
Version	1.0.0
Format	JSON
Prerequisites	• Add an additional mandatory item with category code AVR as a root level item using `hierarchyRelationshipParentItemId`. All other items must be a child of this item. • Link the transportation requirements present in the **Consumer API to Request for Planning in External System** at the item level in the freight document. • For each item that you are performing planning, create an item and pass the `itemBaseBusinessTransactionDocument` and the

Technical Name	Field
	`businessTransactionDocumentReference` with the type code as `LBN_REF`.
Dependencies	Your ordering party has sent the transportation requirements linked to the purchase order or sales order in their Transportation Management on SAP S4/HANA system.
Base Document Type	Type of freight document supported: • 1122 – Transportation Order

You can download the YAML file from the Provider API to Create Planned Freight Document from External System in SAP API Hub.

Data Dictionary

In the SAP API Hub, in the **Provider API to Create Planned Freight Document from External System**, you can view the field details in the **Model View**. In the **API Reference** section, you can view the example values.

In the downloaded YAML file you can view the following details of the fields in the API interface:

- Field name
- Type
- Description
- Max length
- Required or not
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Relevant Codes

SAP Business Network for Logistics has defined its own code values for the various attributes used within business network. These codes are used during B2B communication so using APIs in business network. For more details on the list of codes supported in SAP Business Network for Logistics, see Code Lists.

For planning in external system, the following code lists are relevant:

- **Document Reference Type Codes**

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31	Truck

Code	Description
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42	Isothermic truck
43	Refrigerated truck
46	Refrigerated van
65	Container only vessel

- Unit of Measure Temperature
- Unit of Measure Volume
- Unit of Measure Weight

Additional Information

- **External Document ID**

  Indicates the document identifier in your external system.

- **Check the Status of Messages**

  In the Manage Integration Logs app, you can check the status of the messages. To search for the message, you must enter your LBN ID and the External Document ID in the format `<Carrier LBN ID>-<External Document ID>`. For example, if your **LBN ID** is `1000012` and your **External Document ID** is `FO1001`, then you must enter `100012-F1001` in the search bar.

- **Linking Freight Units to the Freight Document**

  You must link the freight units to the freight document based on the payload in **Consumer API to Request for Planning in External System**. At the item level, you can fine the freight unit identifiers.

- **Base Business Transaction Document**

  For each item that you are performing planning, you must create an item and pass the `itemBaseBusinessTransactionDocument` and `businessTransactionDocumentReference` with type code as `LBN_REF`.

- **FUR Item**

  You can skip the items with the category code as FUR.

- **Cancel Scenario**

  When you perform planning for a freight document with a canceled freight unit, the system generates a freight document in the external transportation system. However, the freight unit does not have a link with the newly created freight document and also items from that freight unit.

- **Cancel Scenario**

  When you perform planning for a freight document with multiple freight units, wherein some of the freight units are canceled and some of them are not canceled, the system generates a freight document in the external transportation system. However, the canceled freight units do not have a link with the newly created freight document and also items from that freight unit.

- **Important Parameters**

The ordering party is a mandatory field. Also, a new field for document type code has been introduced.

Shipper APIs (SOAP APIs)

As a shipper, you can use the APIs in SAP Business Network Freight Collaboration to perform your daily tasks in the areas of freight tendering, freight subcontracting, freight booking, freight tracking, dock appointment scheduling, and freight settlement.

Outsourced Planning

You can create transportation requirements from a sales order, purchase order, or delivery documents in your SAP Transportation Management system and send these transportation requirements to the carrier to request the carrier to perform planning activities in their system. Once the carrier creates a planned freight document in their system, they can send the document containing the transportation requirements back to you.

You can perform the following activities using the APIs for freight settlement:

- You can use the Provider API to Receive Freight Documents for Outsourced Planning to forward a freight document to your SAP Business Network for Logistics system.
- You can use the Consumer API to Send Freight Documents for Outsourced Planning to receive freight documents in your transportation management system.

Provider API to Receive Freight Documents for Outsourced Planning

As a shipper, you can forward a freight document to your SAP Business Network for Logistics system.

Technical Name

`TransportationOrderGenericRequest_Out`

Details

You can use this inbound service provided by SAP Business Network for Logistics to create a central freight document in SAP Business Network for Logistics based on the information from your external system, such as Transportation Management on SAP S/4 HANA. This request will be presented to the external planners.

Structure

For more information, see Request Generic Transportation Order.

Consumer API to Send Freight Documents for Outsourced Planning

As a shipper, you can receive freight documents in your transportation management system.

Technical Name

`TransportationOrderGenericRequest_In`

Details

You can use this outbound service provided by SAP Business Network for Logistics to receive freight document in your external system, such as Transportation Management on SAP S/4 HANA, based on the information provided by SAP Business Network for Logistics. The planned freight document will be sent by your external planner.

Structure

For more information, see Maintain Generic Transportation Order.

Freight Contracting

The freight contracting process comprises of the freight subcontracting and tendering processes.

Tendering

Using the freight tendering APIs, you as a shipper can create, update, cancel, or notify various tendering activities when integrating SAP Business Network for Logistics with external systems.

You can perform the following activities using the APIs for freight tendering:

- You can use the Provider API to Create Freight Request for Quotation create a freight request for quotation in SAP Business Network for Logistics.
- You can use the Provider API to Cancel Freight Request for Quotation to cancel a freight request for quotation in SAP Business Network for Logistics.
- You can use the Provider API to Notify Freight Quotation to notify freight quotation statuses in SAP Business Network for Logistics.
- You can use the Consumer API to Create or Update Freight Quotation to create or update a freight quotation in your transportation management system.

Subcontracting

Using the freight subcontracting APIs, you as a shipper can confirm, create, or cancel various subcontracting activities when integrating SAP Business Network for Logistics with external systems.

You can perform the following activities using the APIs for freight subcontracting:

- You can use the Provider API to Create Freight Order for Subcontracting to create a freight order for subcontracting in SAP Business Network for Logistics.
- You can use the Provider API to Cancel Freight Order for Subcontracting to cancel a freight order for subcontracting in SAP Business Network for Logistics.
- You can use the Consumer API to Confirm Freight Order for Subcontracting to confirm a freight order for subcontracting in your transportation management system.

Provider API to Create Freight Request for Quotation

As a shipper, you can create a freight request for quotation in SAP Business Network for Logistics.

Technical Name

`TransportationOrderQuotationCreateRequest_Out`

Details

You can use this inbound service provided by SAP Business Network for Logistics to create a freight request for quotation from your external system, such as a transportation management system. This request will be presented to your carriers.

Note that SAP Business Network for Logistics accepts only 700 items in a freight request for quotation.

Structure

For more information, see: Request Transportation Order Quotation Creation

Provider API to Cancel Freight Request for Quotation

As a shipper, you can cancel a freight request for quotation in SAP Business Network for Logistics.

Technical Name

`TransportationOrderQuotationCancellationRequest_Out`

Description

You can use this inbound service provided by SAP Business Network for Logistics to cancel a freight request for quotation from your external system, such as a transportation management system. This request will be presented to your carriers.

Structure

For more information, see: Request Transportation Order Quotation Cancellation

Consumer API to Create or Update Freight Quotation

As a shipper, you can create or update a freight quotation in your transportation management system.

Technical Name

`TransportationOrderQuotationConfirmation_In`

Description

You can use this outbound service consumed by SAP Business Network for Logistics to create or update a freight quotation in your external system, such as a transportation management system. The freight quotations are submitted by your carrier.

Structure

For more information, see: Change Transportation Order Based on Transportation Order Quotation Confirmation

Provider API to Notify Freight Quotation

As a shipper, you can notify freight quotation statuses in SAP Business Network for Logistics.

Technical Name

`TransportationOrderQuotationNotification_Out`

Description

You can use this inbound service provided by SAP Business Network for Logistics to notify the status of the freight quotation from your external system, such as a transportation management system.

Structure

For more information, see: Notify of Transportation Order Quotation

Provider API to Create Freight Order for Subcontracting

As a shipper, you can create a freight order for subcontracting in SAP Business Network for Logistics.

Technical Name

`TransportationOrderRequest_Out`

Description

You can use this inbound service provided by SAP Business Network for Logistics to create a freight order for subcontracting from your external system, such as a transportation management system. This request will be presented to your carriers.

Note that SAP Business Network for Logistics accepts only 700 items in a freight order document.

Structure

For more information, see: Request Transportation Order

Provider API to Cancel Freight Order for Subcontracting

As a shipper, you can cancel a freight order for subcontracting in SAP Business Network for Logistics.

Technical Name

`TransportationOrderCancellationRequest_Out`